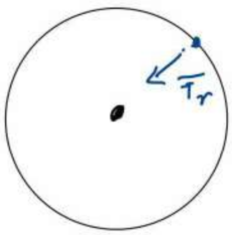
Bohr's model

- Energies are discrete, and correspond to closed orbits with $\oint \underline{p} \cdot d\underline{r} = h \cdot n$
  $(\sim \; L = n \cdot \hbar)$
- These states are stable, do not radiate
- radiation: $E_2 \to E_1 \qquad hf = E_2 - E_1$

Hydrogen spectrum:



$$F_r = \frac{e^2}{4\pi\epsilon_0}\frac{1}{r^2} = m \cdot r\,\omega^2$$

$$E = \frac{1}{2} m r^2 \omega^2 - \frac{\alpha}{r} = -\frac{\alpha}{2 \cdot r} = -\frac{p^2}{2m}$$

$$p = m \cdot r \cdot \omega = \sqrt{2m|E|}$$

$$r = \frac{\alpha}{2|E|}$$

$$\hbar \cdot n = p \cdot r = \alpha \cdot \sqrt{\frac{m}{2}} \cdot \frac{1}{|E|^{1/2}}$$

$$\Rightarrow \quad |E| = \frac{1}{n^2}\frac{1}{\hbar^2}\alpha^2\frac{m}{2} = E_0\frac{1}{n^2}$$

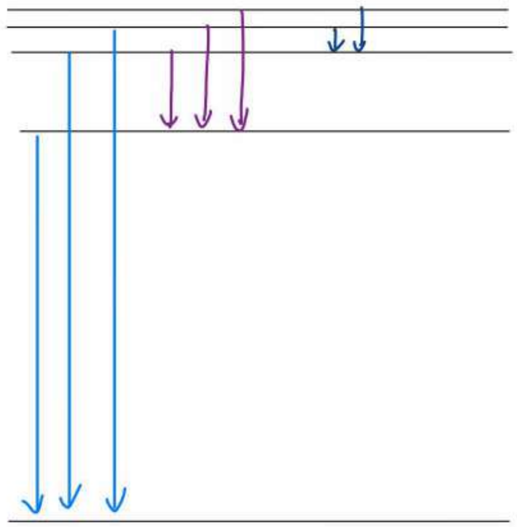
$$\boxed{E_n = -\frac{E_0}{n^2}} \qquad \boxed{E_0 = \frac{m e^4}{2\hbar^2 (4\pi\epsilon_0)^2} = 13{,}6 \text{ eV}}$$

$$= 1 \text{ Ry}$$

1 Rydberg = 13,6 eV

Transitions:

$$\Delta E = h \cdot f = E_0 \left(\frac{1}{n^2} - \frac{1}{k^2} \right)$$



Balmer

Paschen

Lyman

Bohr radious: $E_0 = \dfrac{e^2}{2(4\pi\epsilon_0) a_0} = \dfrac{e^4 \cdot m}{2 \hbar^2 (4\pi\epsilon_0)^2}$

$$\Rightarrow \quad a_B = \frac{\hbar^2 \, 4\pi\epsilon_0}{m \, e^2} = 0{,}53 \, \mathring{A} = 0{,}53 \cdot 10^{-10} m$$

radius of orbit $(n = 1)$

$$\Rightarrow \quad r_n = a_B \cdot n^2$$

($\Rightarrow$ Rydberg atoms !)
$n \gg 1$

velocity ?

$$\frac{m v^2}{2} = E_0 = \frac{m e^4}{2 \hbar^2 (4\pi\epsilon_0)^2}$$

$$v_1 = \frac{e^2}{\hbar \cdot (4\pi\epsilon_0)} \quad \Rightarrow \quad \frac{v_1}{c} \equiv \alpha \cong \frac{1}{137}$$

fine

structure constant

(~ relativistic corrections ...)

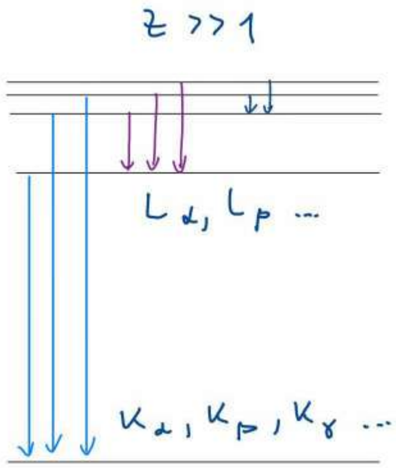
Bohr model:

- predicts spectra of Li^{++} and He^{+}

$\leftrightarrow$ fails for molecules ...

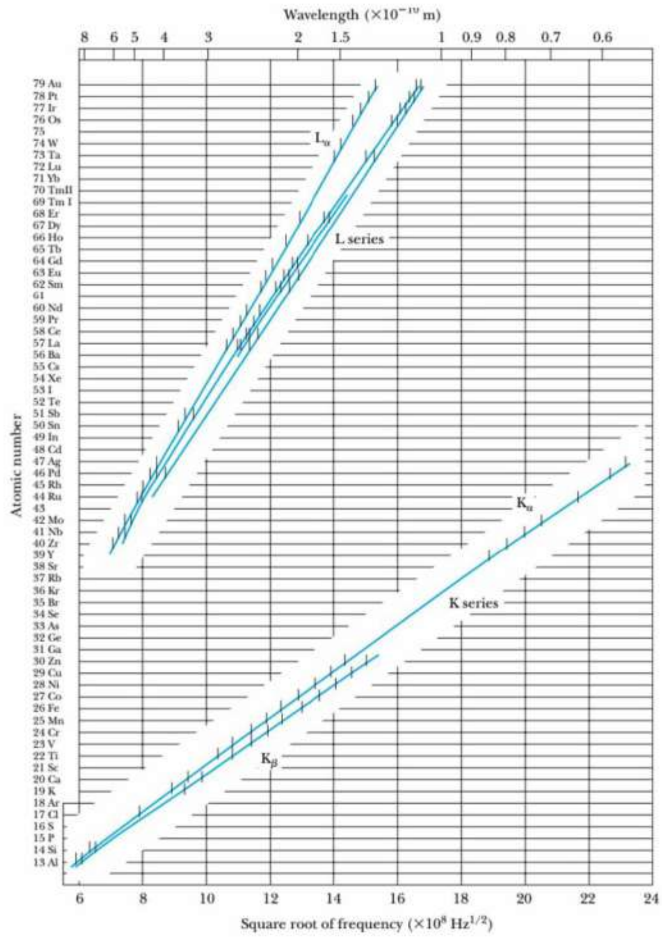
Xray emission lines

Can be extended to X-ray from deep shells:



$M_\alpha, M_\beta, \dots$

$\swarrow$ e^- kicked out $\rightarrow$ K_α transition



$$h \cdot f_K = (Z-1)^2 \left(1 - \frac{1}{n^2}\right)$$

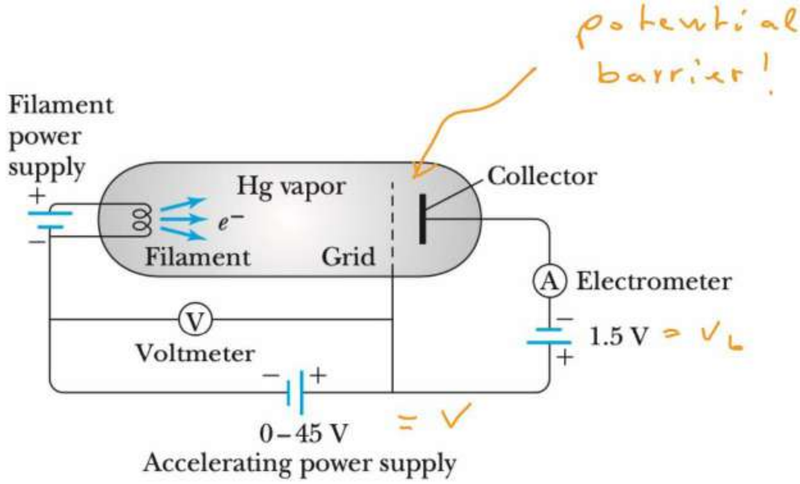
$$h \cdot f_L = Z_{eff}^2 \left(\frac{1}{4} - \frac{1}{n^2}\right)$$

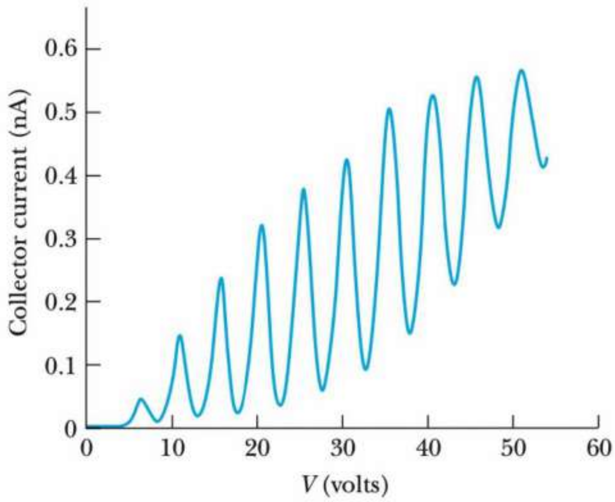
$$Z_{eff,L} \approx Z - 7{,}4$$

core electrons!

Frank-Hertz experiment

Mechanical excitation of atoms





Hg: excited state at $\Delta E = 4.88$ eV

electrons lose energy in discrete units, ΔE !

inelastic collisions slow down electrons:

$$K_{grid} \simeq eV - n \cdot \Delta E$$

n: # of inelastic collisions

kinetic energy at grid, K_{grid} must be sufficient to climb over barrier

$$K_{grid} \overset{!}{\gtrsim} eV_b$$

Matter waves

X-ray diffraction

Compton : X-rays behave as particles (photons)

Q: do they behave as waves, too?

A: Laue, Bragg + Bragg : YES

$E = h \cdot f = 500 \text{ keV} \quad \Rightarrow$

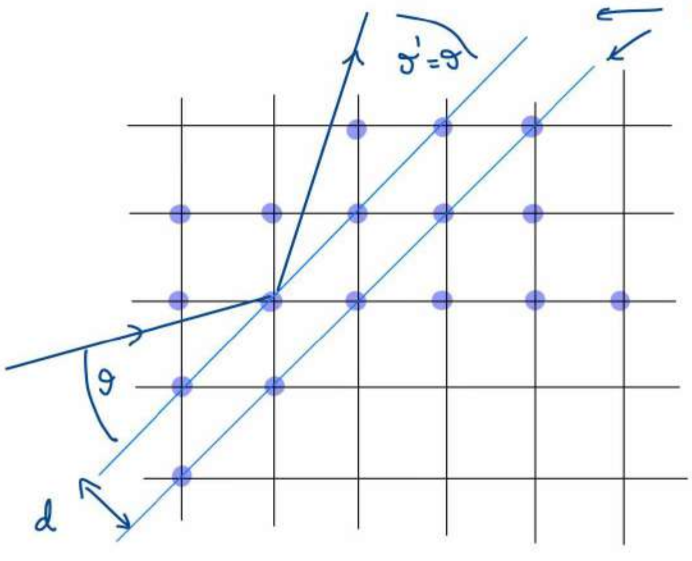
$$\lambda = \frac{c}{f} = \frac{c \cdot h}{30 \text{ keV}} = \frac{1{,}98 \cdot 10^{-25}}{3 \cdot 10^{4} \cdot 1{,}6 \cdot 10^{-19}} \text{ m}$$

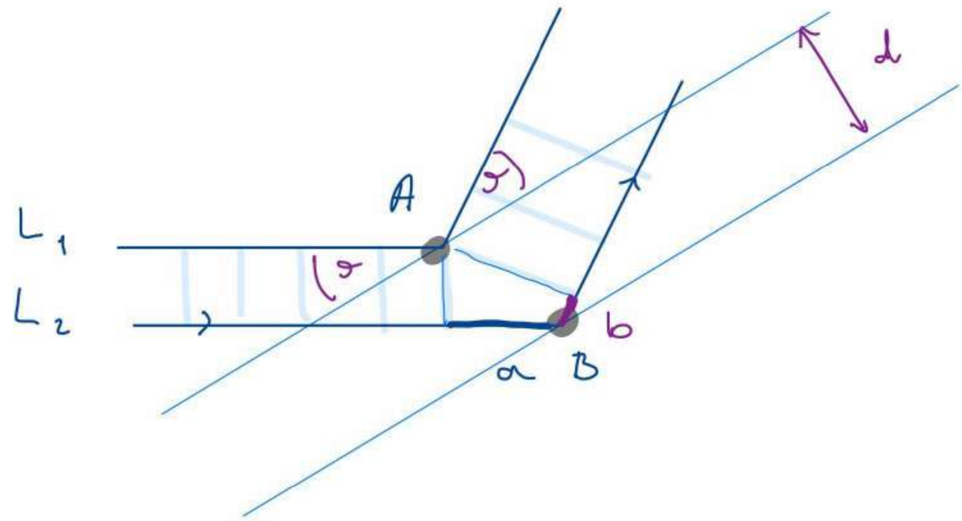
dental

$$= 0{,}41 \cdot 10^{-10} \text{ m}$$

~ atomic distance!

! use crystal lattice for diffraction!





Interference if length difference between P1 and P2 $\quad L_2 - L_1 \overset{!}{=} n \cdot \lambda$

claim:

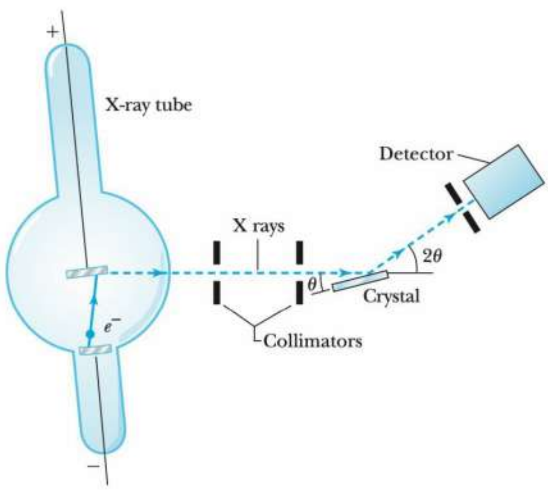
$$L_2 - L_1 = a + b \overset{!}{=} 2d \cdot \sin\vartheta$$

for $\underline{\underline{\text{any}}}$ two atoms in planes

Bragg condition

$$\boxed{2d \sin\vartheta = n \cdot \lambda}$$

Apparatus:



$\Rightarrow$ analysis of complex molecules !

(DNA !)

De Broglie waves

light (wave) can behave as particle

particles should behave as waves

light:

photoeffect: $E = h \cdot f$ (1)

relativity: $p = E/c$ (2)

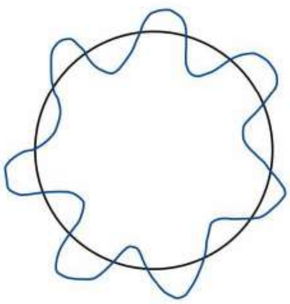
$\Rightarrow$ $\boxed{p = \frac{h \cdot f}{c} = \frac{h}{\lambda} \quad (3)}$ Compton ✓

De Broglie:

particle could also have a wavelength $\boxed{\lambda_e \overset{!}{=} \frac{h}{p_e}}$

Consequences:

- atomic orbit: $2\pi \cdot r = n \cdot \lambda$



$n = 7$

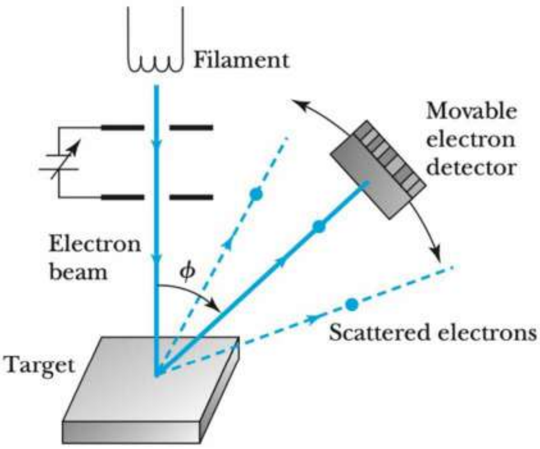
$\Rightarrow p = \frac{h}{\lambda} = n \cdot \frac{h}{2\pi \cdot r}$

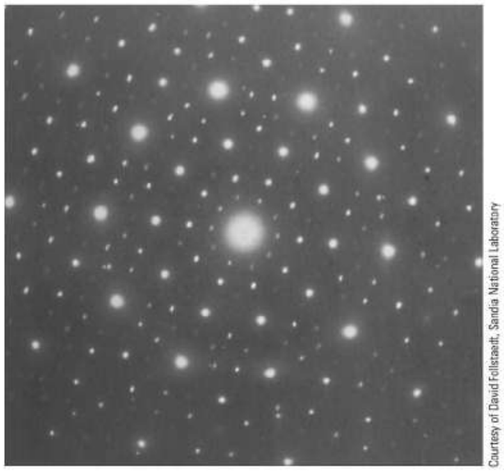
$\Rightarrow L = p \cdot r = n \cdot \hbar$ ✓ !

Bohr !

- electron diffraction

Davisson and Germer



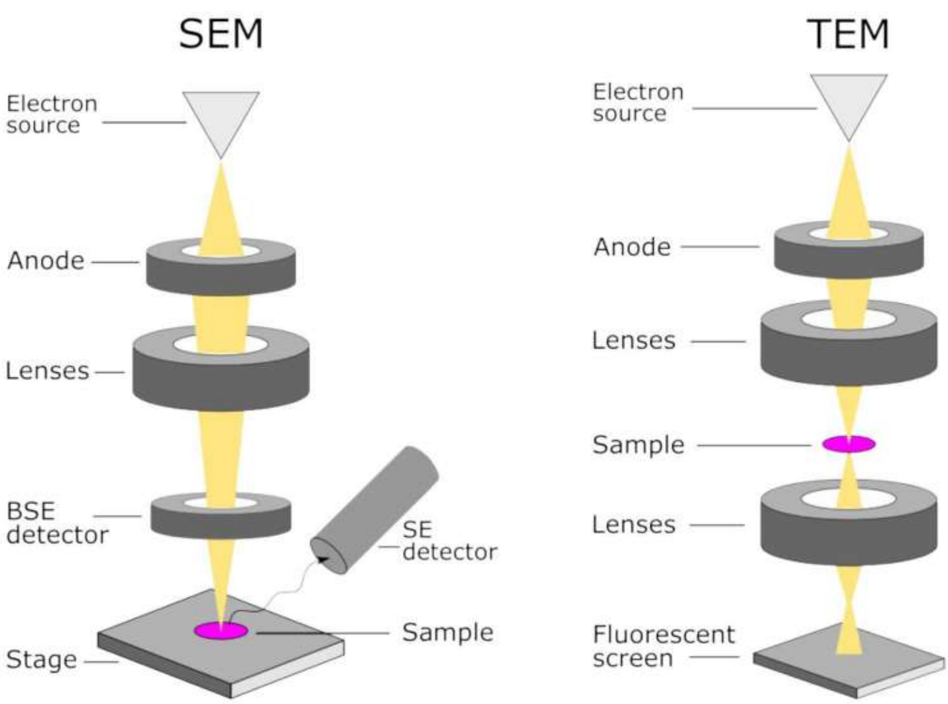


(a)

Transmission electron image !

Devices:
- SEM scanning electron microscope
- TEM transmission e^- μ-scope
- STEM scanning TEM
- neutron diffraction

$\Rightarrow$ resolution $\sim$ 1Å !



Wave motion and group velocity

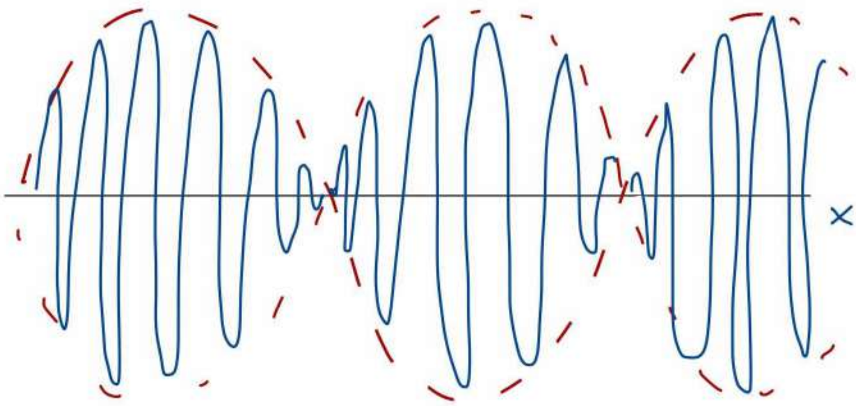
Adding two waves:

$$\psi(x,t) \equiv \cos(k_1 x - \omega_1 t) + \cos(k_2 x - \omega_2 t) =$$

$$k_1 \approx k_2 \qquad \omega_1 \approx \omega_2$$

$$\cos\alpha + \cos\beta = 2\cos\frac{\alpha+\beta}{2}\cos\left(\frac{\alpha-\beta}{2}\right)$$

$$\psi(x,t) = 2\cos(\bar{k}x - \bar{\omega}\cdot t)\cdot\cos\left(\frac{\Delta k}{2}\cdot x - \frac{\Delta\omega}{2}\cdot t\right)$$



- internal pattern moves with velocity

$$v_{ph} = \frac{\bar{\omega}}{\bar{k}}$$ phase velocity

- external curve moves with velocity

$$\boxed{v_g = \frac{\Delta\omega}{\Delta k} = \frac{\partial \omega(k)}{\partial k}}$$ group velocity!

localization in space:

$$\frac{\Delta k}{2} \cdot \Delta x = \pi \Rightarrow \Delta x \simeq \frac{2\pi}{\Delta k}$$

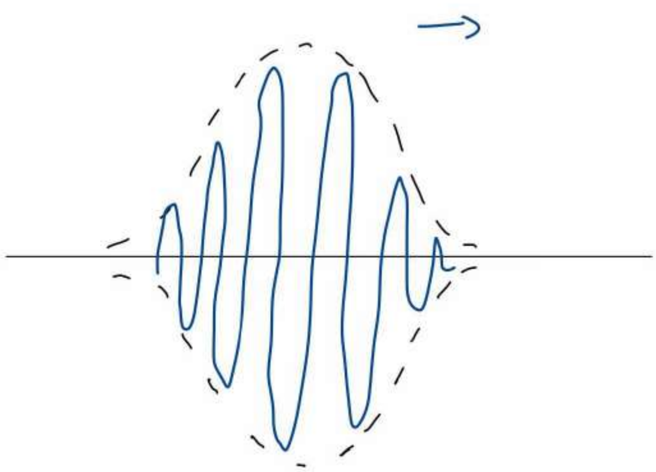
$$p = \frac{h}{\lambda} = \hbar \cdot k \Rightarrow \Delta p = \hbar \cdot \Delta k$$

$$\Rightarrow \boxed{\Delta x \cdot \Delta p \sim h}$$

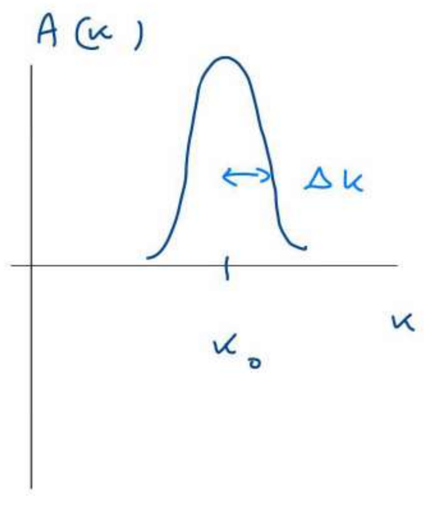
to localize a wave we need many wavelengths!

~ uncertainty

electron : $\psi(x,t) \stackrel{?}{=} \int dk\, A(k) \cos(k \cdot x - \omega(k) \cdot t)$



$\longleftrightarrow \Delta x \sim \frac{1}{\Delta k}$!



" Electron wave :

$$k_0 = \frac{2\pi}{\lambda_0} \rightarrow \text{momentum: } p = \frac{h}{\lambda} = \frac{h}{2\pi} \cdot k_0 = \hbar \cdot k_0$$

$$\text{energy: } \quad E = h \cdot f = \hbar \cdot \omega(k_0) = \frac{p^2}{2 \cdot m} = \frac{\hbar^2}{2m} \cdot k_0^2$$

$$\Rightarrow \quad \boxed{\omega(k) = \frac{\hbar \cdot k^2}{2m}}$$

$$\Rightarrow \quad \psi(x,t) \sim \int dk \; A(k) \; \cos\left(k \cdot x - \frac{\hbar k^2}{2m} \cdot t\right)$$

$$\omega(k) = \frac{\hbar \cdot k^2}{2m}$$

$$\Rightarrow \quad v_g = \left(\frac{\partial \omega}{\partial k}\right)_0 = \frac{\hbar k_0}{m} = p_0 \quad \checkmark$$

$$\Rightarrow \quad v_{ph} = \frac{\omega}{k_0} = \frac{\hbar k_0}{2m} = \frac{p_0}{2m} = \frac{1}{2} v_g \quad !$$

~ fine pattern moves "backwards"

(optical superlattice $\Rightarrow \frac{\partial \omega}{\partial k} \simeq 0 \Rightarrow$ stop / light.)